

14.4 Derivatives of Analytic Functions

As mentioned, a surprising fact is that complex analytic functions have derivatives of all orders. This differs completely from real calculus. Even if a real function is once differentiable we cannot conclude that it is twice differentiable nor that any of its higher derivatives exist. This makes the behavior of complex analytic functions simpler than real functions in this aspect. To prove the surprising fact we use Cauchy's integral formula.

THEOREM 1

Derivatives of an Analytic Function

If $f(z)$ is analytic in a domain D , then it has derivatives of all orders in D , which are then also analytic functions in D . The values of these derivatives at a point z_0 in D are given by the formulas

$$(1') \quad f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz$$

$$(1'') \quad f''(z_0) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^3} dz$$

and in general

$$(1) \quad f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (n = 1, 2, \dots);$$

here C is any simple closed path in D that encloses z_0 and whose full interior belongs to D ; and we integrate counterclockwise around C (Fig. 360).

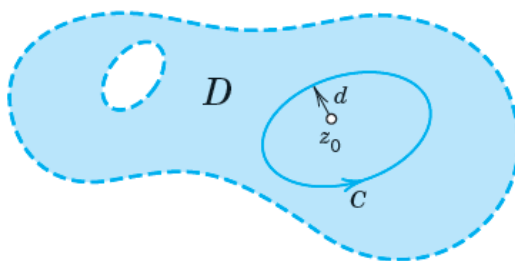


Fig. 360. Theorem 1 and its proof

COMMENT. For memorizing (1), it is useful to observe that these formulas are obtained formally by differentiating the Cauchy formula (1*), Sec. 14.3, under the integral sign with respect to z_0 .

Applications of Theorem 1

EXAMPLE 1 Evaluation of Line Integrals

From (1'), for any contour enclosing the point πi (counterclockwise)

$$\oint_C \frac{\cos z}{(z - \pi i)^2} dz = 2\pi i (\cos z)' \Big|_{z=\pi i} = -2\pi i \sin \pi i = 2\pi \sinh \pi.$$

EXAMPLE 2 From (1''), for any contour enclosing the point $-i$ we obtain by counterclockwise integration

$$\oint_C \frac{z^4 - 3z^2 + 6}{(z + i)^3} dz = \pi i (z^4 - 3z^2 + 6)'' \Big|_{z=-i} = \pi i [12z^2 - 6]_{z=-i} = -18\pi i.$$

EXAMPLE 3 By (1'), for any contour for which 1 lies inside and $\pm 2i$ lie outside (counterclockwise),

$$\begin{aligned} \oint_C \frac{e^z}{(z-1)^2(z^2+4)} dz &= 2\pi i \left(\frac{e^z}{z^2+4} \right)' \Big|_{z=1} \\ &= 2\pi i \frac{e^z(z^2+4) - e^z 2z}{(z^2+4)^2} \Big|_{z=1} = \frac{6e\pi}{25} i \approx 2.050i. \end{aligned}$$

Cauchy's Inequality. Liouville's and Morera's Theorems

We develop other general results about analytic functions, further showing the versatility of Cauchy's integral theorem.

Cauchy's Inequality. Theorem 1 yields a basic inequality that has many applications. To get it, all we have to do is to choose for C in (1) a circle of radius r and center z_0 and apply the ML -inequality (Sec. 14.1); with $|f(z)| \leq M$ on C we obtain from (1)

$$|f^{(n)}(z_0)| = \frac{n!}{2\pi} \left| \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz \right| \leq \frac{n!}{2\pi} M \frac{1}{r^{n+1}} 2\pi r.$$

This gives **Cauchy's inequality**

$$(2) \quad |f^{(n)}(z_0)| \leq \frac{n!M}{r^n}.$$

To gain a first impression of the importance of this inequality, let us prove a famous theorem on entire functions (definition in Sec. 13.5). (For Liouville, see Sec. 11.5.)

THEOREM 2

Liouville's Theorem

If an entire function is bounded in absolute value in the whole complex plane, then this function must be a constant.

THEOREM 3

Morera's² Theorem (Converse of Cauchy's Integral Theorem)

If $f(z)$ is continuous in a simply connected domain D and if

$$(3) \quad \oint_C f(z) \, dz = 0$$

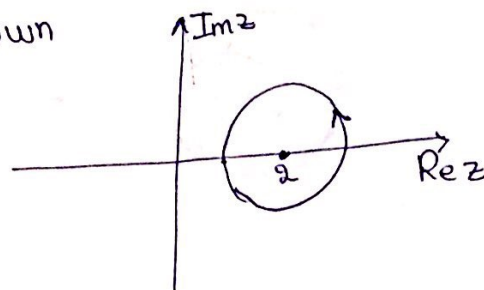
for every closed path in D , then $f(z)$ is analytic in D .

Recommended Exercises 14.3 and 14.4

Examples:

Example #1 Compute $\int_C \frac{e^{z^2}}{z-2} dz$, where C is

the curve shown

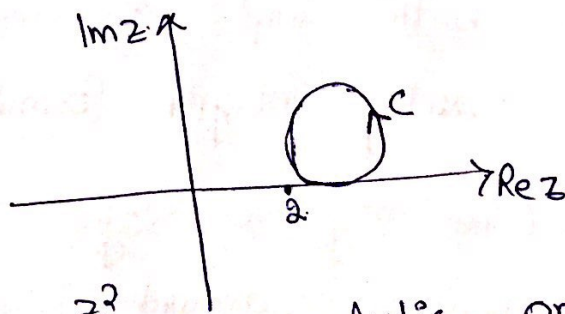


Solution:-

let $f(z) = e^{z^2}$ As $f(z)$ is entire. Since C is the simple closed curve (counterclockwise) and $z=2$ is inside C , so Cauchy's integral formula says that

$$\int_C \frac{e^{z^2}}{z-2} dz = 2\pi i f(2) = 2\pi i e^4 \text{ - Ans.}$$

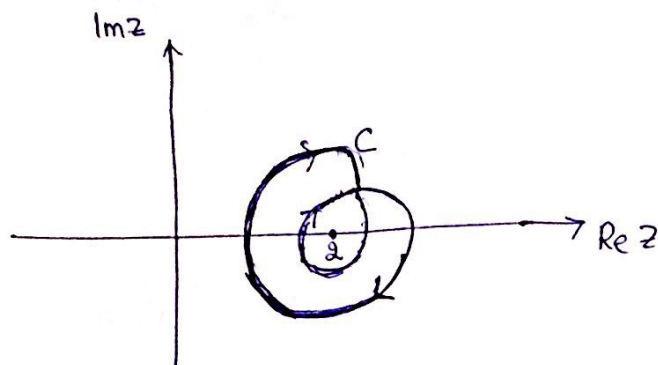
Example #2:- Do the same integral as the previous example with C the curve shown.



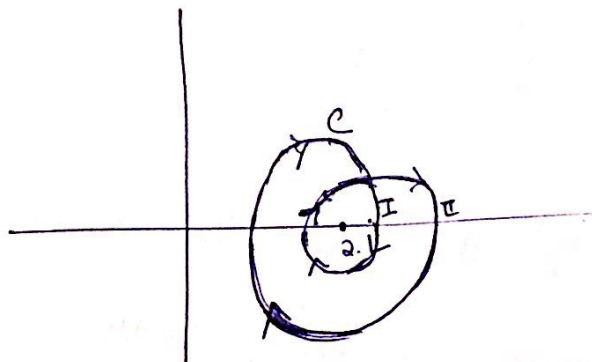
Sol:-

Since $f(z) = \frac{e^{z^2}}{z-2}$ is analytic on and inside C , Cauchy's theorem says that the integral is zero.

Example:- Do the same integral as the previous examples with C the curve shown.



Sol: This one is trickier. Let $f(z) = e^{z^2}$. The curve C goes around a twice in the clockwise direction, so we break C into $C_1 + C_2$ as shown in next figure.



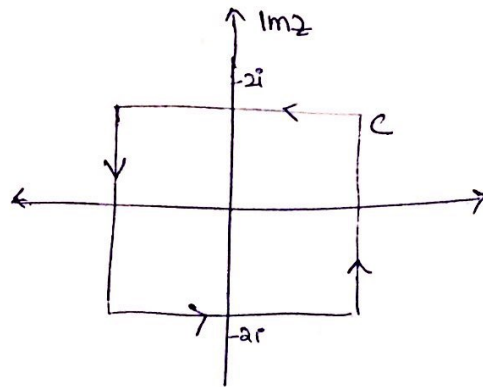
These are both simple so we can apply the Cauchy integral formula to each separately. (The negative signs are because they go clockwise around $z=a$).

$$\int_C \frac{f(z)}{z-a} dz = \int_{C_1} \frac{f(z)}{z-a} dz + \int_{C_2} \frac{f(z)}{z-a} dz = -2\pi i f(a) - 2\pi i f(a) \\ \Rightarrow -4\pi i f(a) = -4\pi i e^{a^2} \text{ Ans.}$$

Ex

Compute $\int_C \frac{\cos z}{z(z^2+8)} dz$ over the contour

Shown.

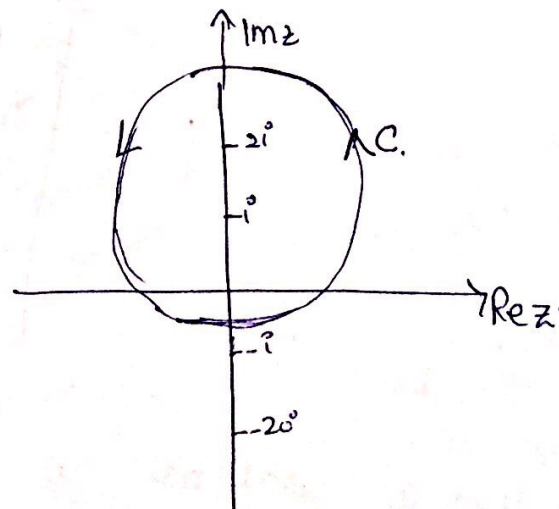


Solution:- let $f(z) = \frac{\cos z}{z^2+8}$, $f(z)$ is analytic on and inside the curve C . That is, the roots of z^2+8 are outside the curve, so we write the integral as

$$\int_C \frac{\left(\frac{\cos z}{z^2+8}\right)}{z} dz = \int_C \frac{f(z)}{z} dz = 2\pi i f(0) = 2\pi i \times \frac{1}{8} = \frac{\pi i}{4}.$$

Example:- Compute $\int_C \frac{1}{(z^2+4)^2} dz$ over the

contour shown.



Solution:- We factor the denominator as

$$\frac{1}{(z^2+4)^2} = \frac{1}{(z-2i)^2(z+2i)^2}$$

let $f(z) = \frac{1}{(z+2i)^2}$

clearly $f(z)$ is analytic inside C , so by

Cauchy's formula for derivatives:

$$\int_C \frac{1}{(z^2+4)^2} dz = \int_C \frac{f(z)}{(z-2i)^2} dz = 2\pi i f'(2i)$$

$$\Rightarrow 2\pi i \left[\frac{-2}{(z+2i)^3} \right]_{z=2i} = \frac{4\pi i}{64i} = \frac{\pi}{16}$$

Example:- $\oint_C \frac{\sin z}{z^2 - 2iz} dz$, C consists of

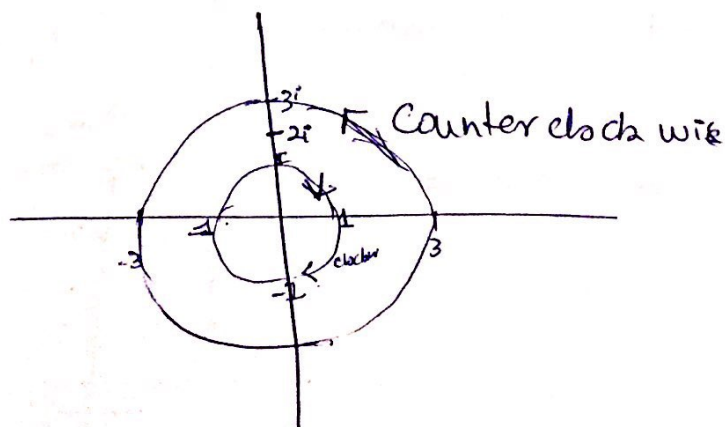
$|z|=3$ (counterclockwise) and $|z|=1$ (clockwise)

Sol:-

$$\oint_C \frac{\sin z}{z(z-2i)} dz$$

$$\Rightarrow \oint_C \frac{\frac{\sin z}{z}}{z-2i} dz$$

As $f(z)$ is analytic on C_1 and C_2



and in a ring shaped domain bounded by C_1 and C_2 and $z_0 = 2i$ is any point in that domain, then.

$$\oint_{C_1} \frac{f(z)}{z - z_0} dz + \oint_{C_2} \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0) \rightarrow (*)$$

$$\oint_C \frac{\sin z}{z(z-2i)} dz = 2\pi i \frac{\sin(2i)}{2i} \Rightarrow \pi \sin(2i)$$

$$\Rightarrow i\pi \sinh 2$$

Power Series, Taylor Series

In Chapter 14, we evaluated complex integrals directly by using Cauchy's integral formula, which was derived from the famous Cauchy integral theorem. We now shift from the approach of Cauchy and Goursat to another approach of evaluating complex integrals, that is, evaluating them by residue integration. This approach, discussed in Chapter 16, first requires a thorough understanding of power series and, in particular, Taylor series. (To develop the theory of residue integration, we still use Cauchy's integral theorem!)

In this chapter, we focus on complex power series and in particular Taylor series. They are analogs of real power series and Taylor series in calculus. Section 15.1 discusses convergence tests for complex series, which are quite similar to those for real series. Thus, if you are familiar with convergence tests from calculus, you may use Sec. 15.1 as a reference section. The main results of this chapter are that complex power series represent analytic functions, as shown in Sec. 15.3, and that, conversely, every analytic function can be represented by power series, called a Taylor series, as shown in Sec. 15.4. The last section (15.5) on uniform convergence is *optional*.

Prerequisite: Chaps. 13, 14.

Sections that may be omitted in a shorter course: 15.1, 15.5.

References and Answers to Problems: App. 1 Part D, App. 2.

15.1 Sequences, Series, Convergence Tests

The basic concepts for *complex* sequences and series and tests for convergence and divergence are very similar to those concepts in (real) calculus. *Thus if you feel at home with real sequences and series and want to take for granted that the ratio test also holds in complex, skip this section and go to Section 15.2.*

Sequences

The basic definitions are as in calculus. An *infinite sequence* or, briefly, a **sequence**, is obtained by assigning to each positive integer n a number z_n , called a **term** of the sequence, and is written

$$z_1, z_2, \cdots \quad \text{or} \quad \{z_1, z_2, \cdots\} \quad \text{or briefly} \quad \{z_n\}.$$

We may also write $z_0, z_1, \cdots$ or $z_2, z_3, \cdots$ or start with some other integer if convenient.

A **real sequence** is one whose terms are real.

Convergence. A **convergent sequence** $z_1, z_2, \dots$ is one that has a limit c , written

$$\lim_{n \rightarrow \infty} z_n = c \quad \text{or simply} \quad z_n \rightarrow c.$$

By definition of **limit** this means that for every $\epsilon > 0$ we can find an N such that

$$(1) \quad |z_n - c| < \epsilon \quad \text{for all } n > N;$$

geometrically, all terms z_n with $n > N$ lie in the open disk of radius ϵ and center c (Fig. 361) and only finitely many terms do not lie in that disk. [For a *real* sequence, (1) gives an open interval of length 2ϵ and real midpoint c on the real line as shown in Fig. 362.]

A **divergent sequence** is one that does not converge.

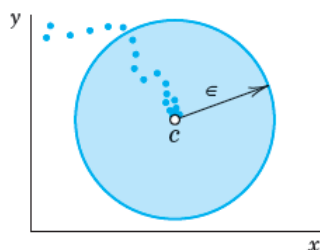


Fig. 361. Convergent complex sequence



Fig. 362. Convergent real sequence

EXAMPLE 1 Convergent and Divergent Sequences

The sequence $\{i^n/n\} = \{i, -\frac{1}{2}, -i/3, \frac{1}{4}, \dots\}$ is convergent with limit 0.

The sequence $\{i^n\} = \{i, -1, -i, 1, \dots\}$ is divergent, and so is $\{z_n\}$ with $z_n = (1 + i)^n$. ■

EXAMPLE 2 Sequences of the Real and the Imaginary Parts

The sequence $\{z_n\}$ with $z_n = x_n + iy_n = 1 - 1/n^2 + i(2 + 4/n)$ is $6i, \frac{3}{4} + 4i, \frac{8}{9} + 10i/3, \frac{15}{16} + 3i, \dots$. (Sketch it.) It converges with the limit $c = 1 + 2i$. Observe that $\{x_n\}$ has the limit $1 = \operatorname{Re} c$ and $\{y_n\}$ has the limit $2 = \operatorname{Im} c$. This is typical. It illustrates the following theorem by which the convergence of a *complex* sequence can be referred back to that of the two *real* sequences of the real parts and the imaginary parts. ■

THEOREM 1

Sequences of the Real and the Imaginary Parts

A sequence $z_1, z_2, \dots, z_n, \dots$ of complex numbers $z_n = x_n + iy_n$ (where $n = 1, 2, \dots$) converges to $c = a + ib$ if and only if the sequence of the real parts $x_1, x_2, \dots$ converges to a and the sequence of the imaginary parts $y_1, y_2, \dots$ converges to b .

Series

Given a sequence $z_1, z_2, \dots, z_m, \dots$, we may form the sequence of the sums

$$s_1 = z_1, \quad s_2 = z_1 + z_2, \quad s_3 = z_1 + z_2 + z_3, \quad \dots$$

and in general

$$(2) \quad s_n = z_1 + z_2 + \dots + z_n \quad (n = 1, 2, \dots).$$

Here s_n is called the ***n*th partial sum** of the *infinite series* or **series**

$$(3) \quad \sum_{m=1}^{\infty} z_m = z_1 + z_2 + \dots.$$

The $z_1, z_2, \dots$ are called the **terms** of the series. (Our usual **summation letter** is n , unless we need n for another purpose, as here, and we then use m as the summation letter.)

A **convergent series** is one whose sequence of partial sums converges, say,

$$\lim_{n \rightarrow \infty} s_n = s. \quad \text{Then we write} \quad s = \sum_{m=1}^{\infty} z_m = z_1 + z_2 + \dots$$

and call s the **sum** or *value* of the series. A series that is not convergent is called a **divergent series**.

If we omit the terms of s_n from (3), there remains

$$(4) \quad R_n = z_{n+1} + z_{n+2} + z_{n+3} + \dots.$$

This is called the **remainder of the series** (3) *after the term* z_n . Clearly, if (3) converges and has the sum s , then

$$s = s_n + R_n, \quad \text{thus} \quad R_n = s - s_n.$$

Now $s_n \rightarrow s$ by the definition of convergence; hence $R_n \rightarrow 0$. In applications, when s is unknown and we compute an approximation s_n of s , then $|R_n|$ is the error, and $R_n \rightarrow 0$ means that we can make $|R_n|$ as small as we please, by choosing n large enough.

An application of Theorem 1 to the partial sums immediately relates the convergence of a complex series to that of the two series of its real parts and of its imaginary parts:

THEOREM 2

Real and Imaginary Parts

A series (3) with $z_m = x_m + iy_m$ converges and has the sum $s = u + iv$ if and only if $x_1 + x_2 + \cdots$ converges and has the sum u and $y_1 + y_2 + \cdots$ converges and has the sum v .

Tests for Convergence and Divergence of Series

Convergence tests in complex are practically the same as in calculus. We apply them before we use a series, to make sure that the series converges.

Divergence can often be shown very simply as follows.

THEOREM 3

Divergence

If a series $z_1 + z_2 + \cdots$ converges, then $\lim_{m \rightarrow \infty} z_m = 0$. Hence if this does not hold, the series diverges.

PROOF If $z_1 + z_2 + \cdots$ converges, with the sum s , then, since $z_m = s_m - s_{m-1}$,

$$\lim_{m \rightarrow \infty} z_m = \lim_{m \rightarrow \infty} (s_m - s_{m-1}) = \lim_{m \rightarrow \infty} s_m - \lim_{m \rightarrow \infty} s_{m-1} = s - s = 0. \quad \blacksquare$$

CAUTION! $z_m \rightarrow 0$ is *necessary* for convergence but *not sufficient*, as we see from the harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$, which satisfies this condition but diverges, as is shown in calculus (see, for example, Ref. [GenRef11] in App. 1).

The practical difficulty in proving convergence is that, in most cases, the sum of a series is unknown. Cauchy overcame this by showing that a series converges if and only if its partial sums eventually get close to each other:

THEOREM 4

Cauchy's Convergence Principle for Series

A series $z_1 + z_2 + \dots$ is convergent if and only if for every given $\epsilon > 0$ (no matter how small) we can find an N (which depends on ϵ , in general) such that

$$(5) \quad |z_{n+1} + z_{n+2} + \dots + z_{n+p}| < \epsilon \quad \text{for every } n > N \text{ and } p = 1, 2, \dots$$

The somewhat involved proof is left optional (see App. 4).

Absolute Convergence. A series $z_1 + z_2 + \dots$ is called **absolutely convergent** if the series of the absolute values of the terms

$$\sum_{m=1}^{\infty} |z_m| = |z_1| + |z_2| + \dots$$

is convergent.

★ If $z_1 + z_2 + \dots$ converges but $|z_1| + |z_2| + \dots$ diverges, then the series $z_1 + z_2 + \dots$ is called, more precisely, **conditionally convergent**.

EXAMPLE 3

A Conditionally Convergent Series

The series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ converges, but only conditionally since the harmonic series diverges, as mentioned above (after Theorem 3). ■

★ If a series is absolutely convergent, it is convergent.

This follows readily from Cauchy's principle (see Prob. 29). This principle also yields the following general convergence test.

Alternating Series test

Suppose that we have series $\sum a_n$ and either $a_n = (-1)^n b_n$ or $a_n = (-1)^{n+1} b_n$ where

$b_n > 0$ for all n , Then if,

(1) $\lim_{n \rightarrow \infty} b_n = 0$ and

(2) $\{b_n\}$ is a decreasing sequence.

Then the series $\sum a_n$ is convergent.

Q Determine if the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

Sol:- here $b_n = 1/n$

1st condition

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

2nd condition

$$b_2 = 1$$

$$b_3 = 1/2$$

$$b_4 = 1/3 \dots \dots \dots$$

Both conditions are satisfied so given series is convergent.

THEOREM 5

Comparison Test

If a series $z_1 + z_2 + \cdots$ is given and we can find a convergent series $b_1 + b_2 + \cdots$ with nonnegative real terms such that $|z_1| \leq b_1, |z_2| \leq b_2, \dots$, then the given series converges, even absolutely.

PROOF By Cauchy's principle, since $b_1 + b_2 + \cdots$ converges, for any given $\epsilon > 0$ we can find an N such that

$$b_{n+1} + \cdots + b_{n+p} < \epsilon \quad \text{for every } n > N \text{ and } p = 1, 2, \dots$$

From this and $|z_1| \leq b_1, |z_2| \leq b_2, \dots$ we conclude that for those n and p ,

$$|z_{n+1}| + \cdots + |z_{n+p}| \leq b_{n+1} + \cdots + b_{n+p} < \epsilon.$$

Hence, again by Cauchy's principle, $|z_1| + |z_2| + \cdots$ converges, so that $z_1 + z_2 + \cdots$ is absolutely convergent. ■

THEOREM 6

Geometric Series

The geometric series

$$(6^*) \quad \sum_{m=0}^{\infty} q^m = 1 + q + q^2 + \cdots$$

converges with the sum $1/(1 - q)$ if $|q| < 1$ and diverges if $|q| \geq 1$.

PROOF If $|q| \geq 1$, then $|q^m| \geq 1$ and Theorem 3 implies divergence.
Now let $|q| < 1$. The n th partial sum is

$$s_n = 1 + q + \cdots + q^n.$$

From this,

$$qs_n = q + \cdots + q^n + q^{n+1}.$$

On subtraction, most terms on the right cancel in pairs, and we are left with

$$s_n - qs_n = (1 - q)s_n = 1 - q^{n+1}.$$

Now $1 - q \neq 0$ since $q \neq 1$, and we may solve for s_n , finding

$$(6) \quad s_n = \frac{1 - q^{n+1}}{1 - q} = \frac{1}{1 - q} - \frac{q^{n+1}}{1 - q}.$$

Since $|q| < 1$, the last term approaches zero as $n \rightarrow \infty$. Hence if $|q| < 1$, the series is convergent and has the sum $1/(1 - q)$. This completes the proof. ■

Ratio Test

This is the most important test in our further work. We get it by taking the geometric series as comparison series $b_1 + b_2 + \cdots$ in Theorem 5:

THEOREM 7

Ratio Test

If a series $z_1 + z_2 + \cdots$ with $z_n \neq 0$ ($n = 1, 2, \dots$) has the property that for every n greater than some N ,

$$(7) \quad \left| \frac{z_{n+1}}{z_n} \right| \leq q < 1 \quad (n > N)$$

(where $q < 1$ is fixed), this series converges absolutely. If for every $n > N$,

$$(8) \quad \left| \frac{z_{n+1}}{z_n} \right| \geq 1 \quad (n > N),$$

the series diverges.

CAUTION! The inequality (7) implies $|z_{n+1}/z_n| < 1$, but this does *not* imply convergence, as we see from the harmonic series, which satisfies $z_{n+1}/z_n = n/(n + 1) < 1$ for all n but diverges.

If the sequence of the ratios in (7) and (8) converges, we get the more convenient

THEOREM 8

Ratio Test

If a series $z_1 + z_2 + \cdots$ with $z_n \neq 0$ ($n = 1, 2, \dots$) is such that $\lim_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| = L$, then:

- (a) If $L < 1$, the series converges absolutely.
- (b) If $L > 1$, the series diverges.
- (c) If $L = 1$, the series may converge or diverge, so that the test fails and permits no conclusion.

EXAMPLE 4

Ratio Test

Is the following series convergent or divergent? (First guess, then calculate.)

$$\sum_{n=0}^{\infty} \frac{(100 + 75i)^n}{n!} = 1 + (100 + 75i) + \frac{1}{2!}(100 + 75i)^2 + \cdots$$

Solution. By Theorem 8, the series is convergent, since

$$\left| \frac{z_{n+1}}{z_n} \right| = \frac{|100 + 75i|^{n+1}/(n+1)!}{|100 + 75i|^n/n!} = \frac{|100 + 75i|}{n+1} = \frac{125}{n+1} \rightarrow L = 0. \quad \blacksquare$$

EXAMPLE 5

Theorem 7 More General Than Theorem 8

Let $a_n = i/2^{3n}$ and $b_n = 1/2^{3n+1}$. Is the following series convergent or divergent?

$$a_0 + b_0 + a_1 + b_1 + \cdots = i + \frac{1}{2} + \frac{i}{8} + \frac{1}{16} + \frac{i}{64} + \frac{1}{128} + \cdots$$

Solution. The ratios of the absolute values of successive terms are $\frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \dots$. Hence convergence follows from Theorem 7. Since the sequence of these ratios has no limit, Theorem 8 is not applicable. $\blacksquare$

Root Test

The ratio test and the root test are the two practically most important tests. The ratio test is usually simpler, but the root test is somewhat more general.

THEOREM 9**Root Test**

If a series $z_1 + z_2 + \cdots$ is such that for every n greater than some N ,

$$(9) \quad \sqrt[n]{|z_n|} \leq q < 1 \quad (n > N)$$

(where $q < 1$ is fixed), this series converges absolutely. If for infinitely many n ,

$$(10) \quad \sqrt[n]{|z_n|} \geq 1,$$

the series diverges.

PROOF If (9) holds, then $|z_n| \leq q^n < 1$ for all $n > N$. Hence the series $|z_1| + |z_2| + \cdots$ converges by comparison with the geometric series, so that the series $z_1 + z_2 + \cdots$ converges absolutely. If (10) holds, then $|z_n| \geq 1$ for infinitely many n . Divergence of $z_1 + z_2 + \cdots$ now follows from Theorem 3. ■

CAUTION! Equation (9) implies $\sqrt[n]{|z_n|} < 1$, but this does not imply convergence, as we see from the harmonic series, which satisfies $\sqrt[n]{1/n} < 1$ (for $n > 1$) but diverges.

If the sequence of the roots in (9) and (10) converges, we more conveniently have

THEOREM 10**Root Test**

If a series $z_1 + z_2 + \cdots$ is such that $\lim_{n \rightarrow \infty} \sqrt[n]{|z_n|} = L$, then:

- (a) The series converges absolutely if $L < 1$.
- (b) The series diverges if $L > 1$.
- (c) If $L = 1$, the test fails; that is, no conclusion is possible.